Evaluation contracts: losses, capabilities, contexts and failures

An evaluation name is not a protocol

“GSM8K”, “MRCR” or “Paloma” leaves many decisions unspecified: dataset revision/split, prompt, few-shot examples, tokenizer/chat template, context/windowing, generation budget, sampling, extractor, metric, routing mode and treatment of failures. Freeze those choices and their resolved values before ranking models.

The Marin evaluation manual anchors actual configurations and records. This page provides the contract needed to compare runs from a different trainer or dataset extension.

Loss sums and denominators

Let $\ell_i = -\log p_\theta(x_i \mid x_{<i})$ and $m_i \in \{0, 1\}$ be the declared score mask. Record sums, not just averages:

$$S_\ell = \sum_i m_i \ell_i, \quad S_t = \sum_i m_i, \quad \mathrm{NLL}_{token} = S_\ell / S_t.$$

For the compatible scored UTF-8 byte count S_b,

$$\mathrm{BPB} = S_\ell / (S_b \ln 2).$$

EOS, padding, prompt-only tokens, overlapping windows and document boundaries affect the mask and byte attribution. A tokenizer change requires a new cache identity and a verified mapping to the same raw text before interpreting cross-tokenizer BPB. Router regularization is a training diagnostic, not part of next-token evaluation NLL.

A pooled corpus estimate sums numerators/denominators across subsets. An equal-subset macro averages the declared subset values. They answer different target-distribution questions. The fixture uses loss sums100/900 and token counts100/300: pooled NLL2.5, macro NLL2.0. Claim.

```
python scripts/kb_workbench.py evaluation-denominators
```

The receipt is synthetic. It is a regression fixture for denominator semantics, not a language-model score.

Keep evaluation cohorts identifiable

Cohort	Purpose	Essential control
Native fixed-distribution loss	Compare recipe/mixture trajectories	Same raw text, tokenizer/cache, packing length and routing/drop mode
Benchmark likelihood	Continuous signals when tiny-model accuracy is near chance	Same choices/answers, shots, extraction and normalization
Generated-answer capability	Measure usable answers	Same template, decoding, token budget, extractor and verifier
Long-context retrieval/reasoning	Test access to distant information	Supported context, length/needle cells, truncation policy and coverage

Cohort	Purpose	Essential control
Agent/environment tasks	Measure task completion with tools	Same environment/image/agent budget, verifier, retries and infrastructure policy

A54-task HF macro, a16-subset Paloma macro and a20-response-task surrogate objective are separate cohorts. A log-probability GSM8K task is not generated-answer GSM8K accuracy. Do not merge these under a shared dashboard label.

Context changes require two curves

When training context increases, retain validation at the old fixed length for comparability and add separately named longer-context evaluation. More conditioning can improve likelihood even without a better model. The context-switch proposal preserves a canonical4K series; multi-context proposal10711 adds explicit length-scoped metrics. Their retrieved PR state is not proof every GPU path was rehearsed.

For served generation, require

$$S_{prompt} + S_{generation} \leq S_{served}.$$

Use the served model's tokenizer to enforce the gate, even if the benchmark bins examples under another official tokenizer. Do not left-truncate a long-context prompt silently and report the score as if the distant information remained available. Publish supported/attempted/scored counts per length cell.

MRCR reference-answer likelihood with full-context versus final-user-only conditioning asks a different question than MRCR generation with a nonce-prefix check and string-similarity scorer. Preserve both definitions. Neither one is a universal long-context capability metric.

Exports are part of the evaluator

Before comparing native loss with served task scores, verify token IDs, padding, special-token rows, chat rendering, representative logits/loss, cache/full-forward agreement and stop-token behavior. A correct weight tensor with a mismatched template can be unusable. Conversely, a new template can improve a score without a change in pretrained weights.

Post-training additionally requires preserving assistant-only loss masks and the weights-only versus optimizer-resume distinction. The generic SFT pipeline's partially green smoke does not qualify a different architecture's tied-embedding optimizer state. Follow the exact model-source branch and recorded validation outcome.

Uncertainty and failure handling

Store per-item predictions and scores, not only aggregate metrics. For paired model comparisons, resample the relevant independent task/example units while preserving pairings. Four attempts from one task are not four independent task samples. Repositories or generated task families may also cluster examples; report that dependency when evaluating cyber or synthetic curricula.

Distinguish incorrect answers, extraction failures, unsupported context, environment errors and model-serving failures. Decide which enter the denominator and publish coverage. Marin's concrete record contract marks infrastructure-error runs below90%coverage as `infra_failed`; adopting that exact threshold is an implementation choice, not a universal statistical rule.

Minimum result receipt

Record checkpoint/HF SHA; consumed tokens and training lineage; resolved evaluation/runtime commit; data revision/split/checksum; prompts/masks/template; shot/seed settings; context/generation/decoding; task-level score sums/denominators; predictions/traces; hardware and failure coverage. Keep base, mid-trained, SFT and RL model roles distinct.

For a cyber extension, create an independent task version with held-out repositories, time ranges or families as appropriate. Benchmark decontamination, verifier correctness and environment determinism must be tested before using scores to select a mixture or synthetic curriculum. Those tasks and acceptance thresholds remain project inputs, not achievements of this repository.